

1. Least squares estimator

$$B = (X^T X)^{-1} X^T Y$$

2. The sum of squared residuals (SSR)

$$S_r = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

3. The sum of squared residuals about the mean (SST)

$$S_t = \sum_{i=1}^m (y_i - \bar{y})^2$$

4. The coefficient of determination (r^2)

$$r^2 = \frac{S_t - S_r}{S_t}$$

5. Scoop Installation (via Windows PowerShell):

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser  
Invoke-RestMethod -Uri https://get.scoop.sh | Invoke-Expression
```

6. Python Installation (via Scoop):

```
scoop install python
```

7. Python Versions

```
python --version
```

8. Installation of Python Packages

```
pip install jupyter numpy matplotlib
```

9. Running Jupyter

```
jupyter lab
```

```
python -m notebook
```